Table 1: IO tuning on the validation set (network trained with the PET and rigidity losses). Stage 1: learning rate / iterations; stage 2: common scale of the four PET weights relative to the training-derived values. Within each stage, the chosen setting (bold) has the lowest $|\Delta\text{MTV}| + |\Delta\text{TLG}|$ among the candidates within two Dice points of the most accurate one ("Within"). Candidates sorted by Dice; patient means. The rigidity weight was fixed at 0.25 (Sec. **??**).

Candidate	Dice (%)	HD95 (mm)	$\lvert\Delta\text{MTV}\rvert$ (%)	$\lvert\Delta\text{TLG}\rvert$ (%)	Sum (%)	NDV (ppm)	Within
Stage 1: learning rate / iterations							
0.1 / 20	78.7	6.58	10.0	9.8	19.8	15	yes
0.1 / 40	78.6	6.58	10.9	10.6	21.5	13	yes
0.1 / 10	78.1	6.69	7.6	8.4	16.0	17	yes
0.05 / 40	78.0	6.78	6.9	8.1	15.0	29	yes
0.05 / 20	77.8	6.81	6.7	8.0	14.7	29	yes
0.05 / 10	77.0	6.84	5.3	7.0	12.2	34	yes
0.025 / 40	77.0	6.89	4.6	6.2	10.7	12	yes
0.025 / 20	76.7	6.93	4.6	6.3	10.9	12	yes
0.0125 / 40	76.1	6.99	3.9	5.7	9.6	2	no
0.025 / 10	75.7	7.14	4.3	5.5	9.8	18	no
0.0125 / 20	75.6	7.16	3.6	5.1	8.7	2	no
0.0125 / 10	75.3	7.17	3.2	4.6	7.8	3	no
Stage 2: PET weight scale (learning rate 0.025, 40 iterations)							
$\times 0$	79.5	6.61	10.1	11.5	21.6	0	yes
$\times$**0.1**	77.9	6.77	5.6	7.8	13.4	7	yes
$\times 0.25$	77.1	6.86	5.1	6.8	11.9	12	no
$\times 0.5$	77.1	6.87	4.9	6.6	11.5	12	no
$\times 1$	77.1	6.87	5.0	6.7	11.6	12	no
$\times 2$	77.0	6.89	4.9	6.6	11.5	13	no
$\times 4$	77.0	6.89	4.6	6.2	10.8	12	no